Chapter 7. Inference in Multiple Regression I

Gauss–Markov, confidence intervals, general linear hypotheses, residuals

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Throughout, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times (p+1)$ and full column rank. We distinguish two sets of assumptions:

- **(A)** $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n)$, needed for exact t and F inference;
- **(B)** $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I_n$ only, enough for the optimality of least squares.

7.1 The Gauss–Markov theorem

Motivation. Least squares has minimal variance among linear unbiased estimators, without normality.

Theorem 7.1 (Gauss–Markov)

Under (B), among linear unbiased $\tilde{\beta} = A\mathbf{y}$, $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ has the smallest variance: $\operatorname{Var}\tilde{\beta} - \operatorname{Var}\hat{\beta}$ is nonnegative definite; equivalently $\mathbf{c}'\hat{\beta}$ is best for every $\mathbf{c}'\beta$.

Proof [Supplement]. $E(A\mathbf{y}) = AX\beta = \beta$ for all β forces $AX = I$. Write $A = (X'X)^{-1}X' + D$; then $DX = AX - I = 0$. Hence

$$\operatorname{Var}(A\mathbf{y}) = \sigma^2 AA' = \sigma^2\{(X'X)^{-1} + (X'X)^{-1}X'D' + DX(X'X)^{-1} + DD'\} = \sigma^2(X'X)^{-1} + \sigma^2 DD',$$

and DD' is nonnegative definite, with $\mathbf{c}'DD'\mathbf{c} = \|D'\mathbf{c}\|^2$. Equality for all $\mathbf{c}$ holds iff $D = 0$. ■

So $\hat{\beta}$ is the **best linear unbiased estimator** (BLUE), needing only uncorrelated equal-variance errors. If $\operatorname{Var}\epsilon = \sigma^2 V$, $V \neq I$, the BLUE is generalized least squares (Chapter 12).

7.2 Confidence intervals

Motivation. Each coefficient interval describes a predictor's effect holding the others fixed.

Under (A), with $(X'X)^{-1} = (c_{ij})$ and $MSE = SSE/(n-p-1)$ independent of $\hat{\beta}$ (Theorem 6.4):

Theorem 7.2

$100(1-\alpha)\%$ confidence intervals are:

(a) for a coefficient: $\hat{\beta}_j \pm t_{\alpha/2}(n-p-1)\sqrt{c_{jj}MSE}$;

(b) for a linear combination $\mathbf{q}'\beta$: $\mathbf{q}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{q}'(X'X)^{-1}\mathbf{q}\,MSE}$;

(c) for the mean response at $\mathbf{x}$: $\mathbf{x}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{x}'(X'X)^{-1}\mathbf{x}\,MSE}$;

(d) a prediction interval for a new observation at $\mathbf{x}$: $\mathbf{x}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}MSE}$.

With σ^2 known, replace MSE by σ^2 and $t_{\alpha/2}$ by $z_{\alpha/2}$.

Proof. (b) contains (a) and (c) as special cases. $\mathbf{q}'\hat{\beta} \sim N(\mathbf{q}'\beta, \sigma^2\mathbf{q}'(X'X)^{-1}\mathbf{q})$ is independent of $(n-p-1)MSE/\sigma^2 \sim \chi^2(n-p-1)$, so the studentized ratio is $t(n-p-1)$.

(d) adds the variance of the new error. ■

7.3 The general linear hypothesis

Many hypotheses are linear restrictions on β:

$$H_0 : C\beta = \mathbf{m} \quad \text{vs} \quad H_1 : C\beta \neq \mathbf{m},$$

with C a known $k \times (p+1)$ matrix of rank k and $\mathbf{m} \in \mathbb{R}^k$. Examples, for $E(y) = \beta_0 + \beta_1 x_1 + \cdots + \beta_4 x_4$:

- $\beta_1 = \beta_2$ and $\beta_3 = 2\beta_4$: $C = \begin{pmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$, $\mathbf{m} = \mathbf{0}$;
- $\beta_1 = \beta_2 = \beta_3 = \beta_4$: three rows $(0, 1, -1, 0, 0)$, $(0, 0, 1, -1, 0)$, $(0, 0, 0, 1, -1)$;
- $\beta_j = 0$: $C = \mathbf{e}_j'$, the j-th unit row.

There are two equivalent routes: minimize SSE under the restriction, or reparametrize to a reduced model.

7.3.1 Method I: constrained least squares

Theorem 7.3 (Restricted least squares)

The minimizer of $\|\mathbf{y} - X\beta\|^2$ subject to $C\beta = \mathbf{m}$ is

$$\tilde{\beta} = \hat{\beta} - (X'X)^{-1}C'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}),$$

and its residual sum of squares is

$$SSE(\tilde{\beta}) = SSE + Q, \qquad Q = (C\hat{\beta} - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}).$$

Proof [Supplement]. With a Lagrange multiplier vector $\lambda \in \mathbb{R}^k$, minimize $L = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta) + 2\lambda'(C\beta - \mathbf{m})$. Setting derivatives to zero:

$$X'X\tilde{\beta} + C'\lambda = X'\mathbf{y}, \qquad C\tilde{\beta} = \mathbf{m}.$$

The first gives $\tilde{\beta} = \hat{\beta} - (X'X)^{-1}C'\lambda$; the second then gives $\lambda = \{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m})$ (invertible, C of full row rank). The decomposition below shows this is the constrained minimum: for any β,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + (\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta),$$

since $X'(\mathbf{y} - X\hat{\beta}) = \mathbf{0}$. With $\beta = \tilde{\beta}$ and $\hat{\beta} - \tilde{\beta} = (X'X)^{-1}C'\lambda$,

$$(\hat{\beta} - \tilde{\beta})'X'X(\hat{\beta} - \tilde{\beta}) = \lambda'C(X'X)^{-1}C'\lambda = (C\hat{\beta} - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}) = Q.$$

For any feasible β, $(\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta) \geq Q$ (it is the squared distance, in the $X'X$ metric, from $\hat{\beta}$ to the affine set $\{C\beta = \mathbf{m}\}$, attained at $\tilde{\beta}$), so $\tilde{\beta}$ is optimal. ■

Theorem 7.4 (F test of the general linear hypothesis)

Under (A),

$$\frac{Q}{\sigma^2} \sim \chi^2(k, \lambda), \qquad \lambda = \frac{(C\beta - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\beta - \mathbf{m})}{2\sigma^2},$$

Q is independent of SSE, and

$$F_0 = \frac{Q/k}{SSE/(n-p-1)} = \frac{Q}{k \cdot MSE} \sim F(k, n-p-1, \lambda).$$

Under H_0, $\lambda = 0$, and the test rejects when $F_0 > F_\alpha(k, n-p-1)$.

Proof. We have

$$C\hat{\beta} - \mathbf{m} \sim N(C\beta - \mathbf{m},\ \sigma^2 C(X'X)^{-1}C'),$$

and Q is its quadratic form with $W = \{C(X'X)^{-1}C'\}^{-1}$; $W \cdot C(X'X)^{-1}C' = I_k$ is idempotent, so Theorem 5.5 applies. Q depends on $\hat{\beta}$ only, independent of SSE (Theorem 6.4). ■

7.3.2 Method II: full and reduced models

Substitute the restriction into the model to get a **reduced model** with $p+1-k$ free parameters; fit it and compute

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/k}{SSE(F)/(n-p-1)},$$

with $df_F = n-p-1$, $df_R = n-p-1+k$ (a nonzero $\mathbf{m}$ becomes an offset). Since $SSE(R) = SSE(\tilde{\beta})$, Theorem 7.3 gives $SSE(R) - SSE(F) = Q$: *both methods give the same statistic.*

For a model without intercept, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times p$, the same holds with $n-p$ in place of $n-p-1$.

Example 7.1

A no-intercept model $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$ is fitted to

x_1	1	2	0	1	3
x_2	0	1	1	2	1
y	3	7	2	5	10

Test $H_0 : \beta_1 = 2\beta_2$. Here

$$X'X = \begin{pmatrix} 15 & 7 \\ 7 & 7 \end{pmatrix}, \qquad X'\mathbf{y} = \begin{pmatrix} 52 \\ 29 \end{pmatrix}, \qquad \hat{\beta} = \frac{1}{56}\begin{pmatrix} 7 & -7 \\ -7 & 15 \end{pmatrix}\begin{pmatrix} 52 \\ 29 \end{pmatrix} = \begin{pmatrix} 2.875 \\ 1.268 \end{pmatrix},$$

$\mathbf{y}'\mathbf{y} = 187$, and $SSE = 187 - \hat{\beta}'X'\mathbf{y} = 0.732$ with 3 degrees of freedom, so $MSE = 0.244$. *Method I*: $C = (1, -2)$, $C\hat{\beta} = 0.339$, $C(X'X)^{-1}C' = \frac{7+28+60}{56} = 1.696$, so $Q = 0.339^2/1.696 = 0.068$ and $F_0 = 0.068/0.244 = 0.28$. *Method II*: under H_0, $y = \beta_2(2x_1 + x_2) + \epsilon$; with $z = 2x_1 + x_2 = (2, 5, 1, 4, 7)$, $\sum z^2 = 95$ and $\sum zy = 133$, so $SSE(R) = 187 - 133^2/95 = 0.800$ and $\{SSE(R) - SSE\}/MSE = 0.068/0.244 = 0.28$. Both give $F_0 = 0.28 < F_{0.05}(1, 3) = 10.13$: H_0 is not rejected.

7.4 Properties of residuals

Motivation. Residuals are correlated with unequal variances $\sigma^2(1 - h_{ii})$, motivating Chapter 10.

Theorem 7.5

For the fitted model with intercept:

(a) $\sum_i e_i = 0$, $\sum_i x_{ij} e_i = 0$ for each j, and $\sum_i \hat{y}_i e_i = 0$;

(b) under (A), $\mathbf{e} = (I - H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I - H))$; in particular $\operatorname{Var} e_i = \sigma^2(1 - h_{ii})$ and $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$.

Proof. (a) is $X'\mathbf{e} = \mathbf{0}$ together with $\hat{\mathbf{y}}'\mathbf{e} = \hat{\beta}' X'\mathbf{e}$.

(b) $E\mathbf{e} = (I - H)X\beta = \mathbf{0}$ and $\operatorname{Var} \mathbf{e} = \sigma^2(I - H)(I - H)' = \sigma^2(I - H)$. ■

So residuals have correlation $-h_{ij}/\sqrt{(1 - h_{ii})(1 - h_{jj})}$ and design-dependent variances. Basic checks plot them against $\hat{y}_i$, each x_{ij} and time order:

- a horizontal band: no problem;
- a funnel: nonconstant variance; use weighted least squares or transform y;
- a linear trend against x_j: a computational error, since (a) forces zero correlation;
- a curve: add a quadratic term (or other function) of x_j, or transform.

7.5 Selected exercises

Motivation. The general linear hypothesis for a constrained mean and for weights summing to one, and the bias–variance effect of a restriction.

Exercise 7.1 (An F statistic under a side condition)

Let $\mathbf{y} = \theta + \epsilon$ with $\mathbf{y}, \theta, \epsilon \in \mathbb{R}^4$ and $\epsilon \sim N_4(\mathbf{0}, \sigma^2 I_4)$, where θ is known to satisfy $\theta_1 + \theta_2 + \theta_3 + \theta_4 = 0$. Show that the F statistic for $H_0 : \theta_1 = \theta_3$ is

$$F_0 = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2},$$

and state its null distribution.

Solution.

Step 1. (The model as a regression.) $\theta \in \Omega = \{\theta : \mathbf{1}'\theta = 0\}$, a 3-dimensional subspace; with the basis $(1, -1, 0, 0)'$, $(0, 1, -1, 0)'$, $(0, 0, 1, -1)'$ as columns of X, $\mathbf{y} = X\beta + \epsilon$ without intercept, and H_0 is one restriction. Method II (Section 7.3): $F_0 = \{SSE(R) - SSE(F)\}/\{SSE(F)/(4 - 3)\}$; for a mean in subspace L, $SSE = \|\mathbf{y} - P_L\mathbf{y}\|^2$ (Theorem 6.1).

Step 2. (Full model.) Ω is the orthogonal complement of $\mathbf{1}$, so $P_\Omega \mathbf{y} = \mathbf{y} - \bar{y}\mathbf{1}$ and

$$SSE(F) = \|\bar{y}\mathbf{1}\|^2 = 4\bar{y}^2 = \frac{(y_1 + y_2 + y_3 + y_4)^2}{4}.$$

Step 3. (Reduced model.) Let $\mathbf{c} = (1, 0, -1, 0)'$. Then $\mathbf{c} \in \Omega$, and the reduced mean space is $\omega = \{\theta \in \Omega : \mathbf{c}'\theta = 0\}$, the orthogonal complement of $\mathbf{c}$ inside Ω. Hence $\Omega = \omega \oplus \operatorname{span}(\mathbf{c})$ orthogonally and $P_\Omega = P_\omega + P_{\mathbf{c}}$, with $P_{\mathbf{c}}\mathbf{y} = (\mathbf{c}'\mathbf{y}/\|\mathbf{c}\|^2)\mathbf{c}$. Since $\mathbf{y} - P_\Omega \mathbf{y} \perp \Omega \ni P_{\mathbf{c}}\mathbf{y}$, Pythagoras gives

$$SSE(R) = \|\mathbf{y} - P_\Omega \mathbf{y} + P_{\mathbf{c}}\mathbf{y}\|^2 = SSE(F) + \frac{(\mathbf{c}'\mathbf{y})^2}{\|\mathbf{c}\|^2} = SSE(F) + \frac{(y_1 - y_3)^2}{2}.$$

Step 4. Therefore

$$F_0 = \frac{(y_1 - y_3)^2/2}{(y_1 + y_2 + y_3 + y_4)^2/4} = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2},$$

and under H_0, $F_0 \sim F(1,1)$ by Theorem 7.4 (with $n-p$ in place of $n-p-1$ for a model without intercept, here $4-3=1$). ■

Exercise 7.2 (Weights that sum to one)

In a survey, n respondents report scores $X_{i1}, \dots, X_{ip}$ on p items and an overall score Y_i. The model is

$$Y_i = \sum_{j=1}^{p} \omega_j X_{ij} + \epsilon_i, \qquad \sum_{j=1}^{p} \omega_j = 1, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

with X $n \times p$ of full rank, $n > p$ (negative weights allowed). Derive the LSE of ω and an unbiased estimator of σ^2, with their distributions.

Solution. Write $G = (X'X)^{-1}$, $\hat{\omega} = GX'\mathbf{Y}$ (the unconstrained estimator), $\mathbf{g} = G\mathbf{1}$ and $s = \mathbf{1}'G\mathbf{1} > 0$.

Step 1. (Estimator.) Theorem 7.3 with $C = \mathbf{1}'$ ($k=1$) and $\mathbf{m} = 1$ gives

$$\tilde{\omega} = \hat{\omega} - \frac{\mathbf{1}'\hat{\omega} - 1}{s}\mathbf{g} = \Big(I - \frac{\mathbf{g}\mathbf{1}'}{s}\Big)\hat{\omega} + \frac{\mathbf{g}}{s},$$

and indeed $\mathbf{1}'\tilde{\omega} = \mathbf{1}'\hat{\omega} - (\mathbf{1}'\hat{\omega} - 1) = 1$.

Step 2. (Reduced model.) Substituting $\omega_p = 1 - \sum_{j<p} \omega_j$,

$$Y_i - X_{ip} = \sum_{j<p} \omega_j (X_{ij} - X_{ip}) + \epsilon_i,$$

an unconstrained regression of $\mathbf{Y} - \mathbf{X}_p$ on $Z = XB$, B the $p \times (p-1)$ matrix with columns $\mathbf{e}_j - \mathbf{e}_p$; Z has full column rank $p-1$. As $(\omega_1, \dots, \omega_{p-1}) \mapsto \omega$ is a bijection onto the constraint set preserving $\|\mathbf{Y} - X\omega\|^2$, $\tilde{\omega}$ is this regression's estimator ($\tilde{\omega}_p = 1 - \sum_{j<p} \tilde{\omega}_j$), with residual sum of squares $SSE(\tilde{\omega})$ on $n-p+1$ d.f.

Step 3. (σ^2.) Theorem 6.4 for the reduced regression gives $SSE(\tilde{\omega})/\sigma^2 \sim \chi^2(n-p+1)$, independent of $\tilde{\omega}$. Thus

$$\tilde{\sigma}^2 = \frac{\|\mathbf{Y} - X\tilde{\omega}\|^2}{n-p+1}$$

is unbiased, and $(n-p+1)\tilde{\sigma}^2/\sigma^2 \sim \chi^2(n-p+1)$.

Step 4. (Distribution of $\tilde{\omega}$.) $\tilde{\omega}$ is an affine function of $\hat{\omega} \sim N_p(\omega, \sigma^2 G)$, hence normal. Since $\mathbf{1}'\omega = 1$, $E\tilde{\omega} = \omega - \frac{1-1}{s}\mathbf{g} = \omega$. With $M = I - \mathbf{g}\mathbf{1}'/s$,

$$\operatorname{Var}\tilde{\omega} = \sigma^2 MGM' = \sigma^2\Big(G - \frac{\mathbf{g}\mathbf{g}'}{s} - \frac{\mathbf{g}\mathbf{g}'}{s} + \frac{\mathbf{g}(\mathbf{1}'G\mathbf{1})\mathbf{g}'}{s^2}\Big) = \sigma^2\Big(G - \frac{\mathbf{g}\mathbf{g}'}{s}\Big).$$

So $\tilde{\omega} \sim N_p(\omega, \sigma^2(G - \mathbf{g}\mathbf{g}'/s))$, a singular normal distribution concentrated on the hyperplane $\mathbf{1}'\omega = 1$: indeed $(G - \mathbf{g}\mathbf{g}'/s)\mathbf{1} = \mathbf{g} - \mathbf{g} = \mathbf{0}$. Replacing σ^2 by $\tilde{\sigma}^2$ gives t intervals for each ω_j with $n-p+1$ degrees of freedom. ■

Exercise 7.3 (Bias and variance of restricted least squares)

Under assumption (B), let $\tilde{\beta}$ be the restricted estimator of Theorem 7.3, and $G = (X'X)^{-1}$, $W = \{CGC'\}^{-1}$.

(a) Show that $E\tilde{\beta} = \beta - GC'W(C\beta - \mathbf{m})$; in particular $\tilde{\beta}$ is unbiased when the restriction is true.
(b) Show that $\operatorname{Var}\tilde{\beta} = \sigma^2(G - GC'WCG)$, so that $\operatorname{Var}\hat{\beta} - \operatorname{Var}\tilde{\beta}$ is nonnegative definite whether or not the restriction is true.

Solution [Supplement].

(a) By Theorem 7.3, $\tilde{\beta} = M\hat{\beta} + GC'W\mathbf{m}$ with $M = I - GC'WC$. Since $E\hat{\beta} = \beta$,

$$E\tilde{\beta} = \beta - GC'WC\beta + GC'W\mathbf{m} = \beta - GC'W(C\beta - \mathbf{m}),$$

which equals β when $C\beta = \mathbf{m}$.

(b) $\operatorname{Var}\tilde{\beta} = M(\sigma^2 G)M'$. Using $G' = G$, $W' = W$ and $W(CGC')W = W$,

$$\begin{aligned} MGM' &= (G - GC'WCG)(I - C'WCG) \\ &= G - GC'WCG - GC'WCG + GC'W(CGC')WCG = G - GC'WCG. \end{aligned}$$

So $\operatorname{Var}\hat{\beta} - \operatorname{Var}\tilde{\beta} = \sigma^2(CG)'W(CG)$ is nonnegative definite (W positive definite): a correct restriction reduces variance without bias; a false one adds the bias of (a). ■